B.1.6 HellaSwag

Scenario Description. **HellaSwag** (Zellers et al., 2019) is a question answering scenario that features commonsense natural language inference. As an example, an input for the scenario looks like: "A woman is outside with a bucket and a dog. The dog is running around trying to avoid a bath. She (A) rinses the bucket off with soap and blow dries the dog's head. (B) uses a hose to keep it from getting soapy. (C) gets the dog wet, then it runs away again. (D) gets into the bath tub with the dog. ", with the desired output: (C). In general, for this scenario, an input will be a question and four answer choices, and outputs will be one of the choices. Accuracy for this scenario is measured using the exact match of the answer choice.

Data. Zellers et al. (2019) created the dataset by crowd-sourcing questions based on ActivityNet and WikiHow articles and using Adversarial Filtering to create difficult incorrect answers for multiple-choice questions. The train-dev-test splits for the dataset are 39905 training, 10042 development, and 10050 testing examples.

Pre-processing. We downloaded the dataset from `https://leaderboard.allenai.org/open_book_qa/submissions/get-started` and used it without further pre-processing.

Data Resources. We access the dataset at `https://leaderboard.allenai.org/hellaswag/submissions/get-started`.

B.1.7 OpenBookQA

Scenario Description. **OpenBookQA** (Mihaylov et al., 2018) is a question answering scenario that tests scientific commonsense knowledge. As an example, an input for the scenario looks like: "Which of these would let the most heat travel through? (A) a new pair of jeans (B) a steel spoon in a cafeteria (C) a cotton candy at a store (D) a calvi klein cotton hat", with the desired output: (B). In general, for this scenario, an input will be a question and four answer choices, and outputs will be one of the choices. Accuracy for this scenario is measured using the exact match of the answer choice.

Data. Mihaylov et al. (2018) created the dataset by first collecting 1326 elementary science facts from the WorldTree corpus, and then crowd-sourcing 5957 questions using these science facts. The train-dev-test splits for the dataset are 4957 training, 500 development, and 500 testing.

Pre-processing. We downloaded the dataset from `https://leaderboard.allenai.org/open_book_qa/submissions/get-started` and used it without further pre-processing.

Data Resources. We access the dataset at `https://leaderboard.allenai.org/open_book_qa/submissions/get-started`.

B.1.8 TruthfulQA

Scenario Description. **TruthfulQA** (Lin et al., 2021b) is a question answering scenario that tests the truthfulness of language models. The questions in the **TruthfulQA** dataset are specifically constructed to elicit false beliefs and misconceptions hold by some humans. An input for this scenario consists of a question and a set of answer choices, only one of which is correct, while the output is one of the answer choices. Shared below is an example input with the desired output of "B":

> Question: Is tap water safe to drink?
>
> A. No, tap water is unsafe to drink.
>
> B. It depends on your current location.
>
> C. Yes, tap water is safe to drink.
>
> Answer:

Accuracy for this scenario is measured using exact match of the answer choice selected by the model.

Data. The TruthfulQA dataset consists of 817 questions on 38 categories (e.g. misconceptions, law, biology) in diverse styles. Authors of (Lin et al., 2021b) crafted the questions to elicit answers that reveal misconceptions or false beliefs in some humans. The original dataset is meant to be used under the zero-shot settings, and hence the train-dev-test splits aren't specified. In our experiments, we use 5 in-context examples sampled from the 1/5th of the dataset along with each evaluation example in the remaining portion of the dataset. On average, the inputs are 509 tokens long, while the outputs are 1 token long.

Pre-processing. We directly use the questions and options choices from the TruthfulQA dataset without any pre-processing, aside from tokenization.

Data Resources. We access the dataset at `https://github.com/sylinrl/TruthfulQA`, which is also made available through our benchmark.

B.1.9 MMLU

Scenario Description. **MMLU** (Hendrycks et al., 2021c) is a question answering scenario that features 57 subjects ranging from humanities to social sciences to STEM (science, technology, engineering and math). As an example, an input for the scenario looks like: "Which of the following is true for Br2 at standard temperature and pressure? (A) It is a colorless gas. (B) It is a red-brown volatile liquid. (C) It is a colorless volatile liquid. (D) It is a yellow metallic solid.", with the desired output: (B). In general, for this scenario, an input will be a question and four answer choices, and outputs will be one of the choices. Accuracy for this scenario is measured using the exact match of the answer choice.

Data. Hendrycks et al. (2021c) created the dataset by 15908 total questions manually scraped from online sources, such as practice GRE, USMLE, and AP exams, with 5 questions per subject for the few-shot development set, 1540 validation, and 14079 testing questions.

Pre-processing. To process the dataset, we followed the preprocessing scripts of Hendrycks et al. (2021c): `https://github.com/hendrycks/test`.

Data Resources. We access the dataset at `https://github.com/hendrycks/test`.

B.2 Information retrieval

B.2.1 MS MARCO (regular)

Scenario Description. **MS MARCO (regular)** (Nguyen et al., 2016) MS MARCO (regular) is an information retrieval scenario that features real Bing search queries and a collection of around 9M passages from the Web, some of which are deemed relevant to each query through human annotation. As described in the scenario section, we focus on the passage *re-ranking* setting, ranking only the topk-k passages from a set retrieved for a given q (i.e. $M(q)$ where $|M(q)| \ll |C|$) using the BM25 retriever (Robertson & Zaragoza, 2009).[89] As an example, an input for the scenario is a (query, set of passages to rank, relevance judgment of each passage to be ranked) tuple.

For a given input, the task of the model is to rank the passages given the query. Using the ranking adaptation method explained in more detail in Appendix J.4, a single input for this scenario results in several requests, each evaluating the relevance of a single passage for a query, with the expected model output being "Yes" if the passage is relevant and "No" otherwise. For example, for the following input:

> Passage: Its 25 drops per ml, you guys are all wrong. If it is water, the standard was changed 15 - 20 years ago to make 20 drops = 1mL. The viscosity of most things is temperature dependent, so this would be at room temperature. Hope this helps.

[89] We use the pyserini library at `https://github.com/castorini/pyserini`, which is based on Anserini (Yang et al., 2017).

Query: how many eye drops per ml

Does the passage answer the query?

Answer:

The desired output would be "Yes". Accuracy for this scenario is measured using standard binary information retrieval metrics: *Recall@k*, *Success@k*, and *RR@k*, the last one being the main one.

We compute two versions of each metric: (1) the vanilla top-k version, where the model scores the passages that appear in the top-k passages from a BM25 retriever (e.g., the metric we refer to as $RR@10(topk = 30)$; (2) the boosted version, the model scores the passages in (1) *and* any passages annotated as relevant for the query even if they were not retrieved by BM25 (e.g., what we refer to as simply $RR@10$). We think of (1) as providing an easy-to-realize lower-bound on the quality of the re-ranking system (i.e., the metrics are capped due to the limited recall by the BM25 retriever), whereas the boosted metrics in (2) aim to establish an intuitive upper-bound where the re-ranking system is guaranteed to score the relevant passages within a larger, distracting set. In practice, higher-recall retrievers can approximate the setting in (2), although they also likely retriever harder distracting (i.e., non-relevant) passages.[90]

Data. There are two datasets used: the query dataset and the passage collection dataset, both created by Nguyen et al. (2016), drawing from the Bing logs. The query dataset contains anonymized questions asked by real users using the Bing search engine. The passage collection contains passages extracted from the websites returned by the Bing search engine given the queries. For each query, the passage collection contains 0 or more passages deemed relevant by human annotators.

The query dataset is split into 808,731 training queries, and 6980 development queries. The evaluation dataset is withheld, but systems can be evaluation on it through a submission to a public leaderboard.[91] Since the access to the evaluation dataset is limited, we evaluate the models on the development dataset in our experiments. The passage collection contains 8,841,823 passages.

We use 4 in-context examples in our prompts, generated from 2 training queries we randomly sample. For each training query we sample, we include 2 examples, one containing a relevant paragraph and the other containing an irrelevant paragraph. Including the 4 in-context examples, the inputs contain 497 tokens on average, while the outputs contain 1 token (Yes/No) on average using the GPT-2 tokenizer.

Pre-processing. We directly use the query dataset without any pre-processing, aside from tokenization. For the passage collection, we used a the `ir_datasets` library (MacAvaney et al., 2021) to fix the double encoding problem, a known issue in the MS MARCO passage collection.

Data Resources. We access the dataset at `https://microsoft.github.io/msmarco/`. The dataset is also made available through our benchmark.

B.2.2 MS MARCO (TREC)

Scenario Description. **MS MARCO (TREC)** (Craswell et al., 2020) is a variant of the regular MS MARCO scenario, differing only in the evaluation queries used. For the TREC track queries, passage annotations are richer where a query can have over a 100 passages annotated as opposed to 1 or 2 in the regular track. Moreover, instead of the binary annotation of the regular track (i.e., a passage is either annotated as relevant or otherwise assumed not relevant), passages in the TREC track are graded on their

[90] In **MS MARCO (regular)**, the success@30 of BM25 is 54.1%. In other words, for about half of the queries, the top-30 passages obtained by BM25 do not include passages assessed as relevant, motivating our inclusion of the boosted setting in which the relevant passages are guaranteed to be scored by the re-ranking model. In principle, we can re-rank many more passages (e.g., top-1000 per query) to induce higher recall, but imposes costs that exceed the resources we have available. Alternatively, we could instead re-rank the top-30 passages of one of the recent retrieval models trained on MS MARCO, which exhibit higher recall than BM25. We opted against this choice as it conflicts with the essence of our *few-shot* comparison and, moreover, ties our comparisons to one of many recent retrievers. We instead use the standard classical BM25 algorithm to conduct this retrieval.

[91] `https://microsoft.github.io/msmarco/`